Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/01/2025
PI Name:	Iyer, Janani Ph.D.		
Project Title:	Sex-Specific Physiological and Transcriptomic CNS Responses to Combined Effects of Spaceflight Stressors in Drosophila melanogaster		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
Start Date:	05/01/2023	**End Date:**	04/30/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 04/30/2026 per F. Hernandez/ARC. (Ed., 3/28/25). NOTE: End date changed to 04/30/2025 per F. Hernandez/ARC. Previous end date was 3/15/24 (Ed., 3/15/24).		
Key Personnel Changes/Previous PI:	May 2025 Update: Per NASA, Dr. April Ronca, a former Co-Investigator on the grant, recently retired and has left the project. Dr. Joshua Alwood, a senior scientist at NASA Ames Research Center (ARC), will assume the role of NASA Institutional PI/Co-Investigator. Dr. Alwood brings extensive experience in both spaceflight and ground-based animal studies and has led numerous grants involving collaborations across government and academic institutions (Ed., 5/1/2025). FY25 - No changes		
COI Name (Institution):	Mhatre, Siddhita Ph.D. (NASA Ames Research Center) Alwood, Joshua (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	During space exploration, damage to the central nervous system (CNS) due to altered gravity is a significant risk, along with the constant exposure to elevated CO2 levels. The combination of these stressors can negatively impact the CNS health that may lead to decrements in astronaut performance, posing a risk to the crew and the mission. Thus, there is an unmet need to unravel the mechanisms and pathways affected by these combined spaceflight stressors. In this proposal, we aim to address sex-specific and long-term responses to spaceflight stressors (mimicking the longitudinal post-flight evaluations in astronauts). We will perform behavioral, brain morphological, and biochemical assays, along with cell-specific transcriptomic profiling in Drosophila melanogaster to investigate the underlying mechanistic responses to single and combined exposures of altered gravity and elevated CO2. While this solicitation is limited in time and scope, we anticipate that our findings from this study will inform future investigations in vertebrate models and contribute to new research that will address the biomedical outcomes of deep space stressors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The effects of gravity as a continuum on the central nervous system (CNS) are less explored. Also, even though elevated CO2 has been a long-known stressor in spaceflight, data on its effects on different systems, specifically CNS, is very scant. Further, the combined effects of altered gravity and elevated CO2 are unknown. This study will focus on the neuronal responses to oxidative stress induced by altered gravity and elevated CO2. Additionally, the proposal offers a unique opportunity to evaluate key behavioral, biochemical, cellular, morphological, and molecular responses longitudinally over multiple time points post-exposure to the stressors, which are currently not well-characterized. The genetic basis of responses to oxidative stress in Drosophila and vertebrates is conserved, thus this study will serve as the first step in identifying pathways and mechanisms that can be used for therapeutic intervention. Neurological Disorders: Studies suggest that elevated CO2 levels may exacerbate symptoms in individuals with neurological disorders such as Alzheimer's disease and Parkinson's disease. Research in this area helps elucidate the mechanisms through which CO2 affects neurological function and may provide insights into potential therapeutic interventions to mitigate these effects and improve the quality of life for affected individuals.
Task Progress:	Ground-based studies, such as hypergravity (HG) using centrifugation, offer a cost-effective and reliable means to simulate gravitational conditions found on Earth. Previous research utilizing animal spaceflight and HG studies has provided valuable insights into the impact of altered gravity on behavioral and neural functions. This study aims to explore dose-dependent responses to altered gravity by comparing central nervous system (CNS) reactions under two different gravitational loads: 1.2g and 3g. Additionally, we seek to investigate the effects of elevated CO2 levels, which are commonly observed in closed space habitats like the International Space Station (ISS) due to technical challenges in CO2 scrubbing. Recent literature and our preliminary investigations indicate that prolonged exposure to elevated CO2 can influence CNS physiology, immune response, and key metabolic pathways, such as oxidative phosphorylation. Astronauts experience constant exposure to altered gravity (AG) and elevated CO2 levels during spaceflight, yet the combined effects of AG and elevated CO2 on the CNS remain poorly understood. Our study is designed to address this gap by subjecting adult flies, aged 3 days, to chronic HG (1.2g and 3g) and/or elevated CO2 (4000 ppm) for 15 days. The selection of 1.2g and 3g as gravitational loads was based on preliminary findings, with 3g chosen to evaluate CNS responses across a gravity continuum. The CO2 concentration of 4000 ppm reflects average levels observed on the ISS during our MVP-Fly-01 spaceflight mission. Our treatment paradigm allows for the longitudinal study of CNS responses to altered gravity, with data collection at immediate post-treatment (R+0), ten days post-treatment (R+10), and twenty-five days post-treatment (R+25). This comprehensive approach will provide valuable insights into the combined effects of altered gravity and elevated CO2 on CNS function, shedding light on the challenges faced by astronauts during space missions, and informing strategies to mitigate potential adverse effects. The data at R0 indicates there is a significant reduction in climbing response in 1.2g females, 3g (males and females), and 3gCO2 (males and females). A similar response was noted in the MVP-FLY-01 mission immediately after the return from the ISS. Further, following the climbing ability response at R10 day in 1.2g is rescued to 1g levels in both males and females, but 3g and 3gCO2 remain significantly lowered. Comparision of R0 and R25 timepoint data side-by-side for each condition, we observed age-related decrease in both 1g male and female flies, while this effect is exacerbated in 1.2g, 1gCO2, and 1.2gCO2 males. In females, 1gCO2 and 1.2gCO2 showed the most difference in climbing ability between R0 and R25. The data suggest that an individual stressor – specifically, gravity – causes immediate changes in climbing behavior; the behavior at 1.2g is rescued at R10 timepoint, while the dual stressor at 3g+CO2 still shows a continual significant reduction. We assessed the number of dopaminergic (DA) neurons following exposure to HG+CO2. In males, DA neuron counts were significantly reduced across all conditions, compared to the 1g control group. In females, while neuron counts decreased in all conditions except for 1.2g and 1.2gCO2, a downward trend was still apparent. Overall, the data indicate that hypergravity contributes to a decrease in DA neuron counts, with the most notable reduction seen under 3gCO2 conditions in males, and a similar trend observed in females. Furthermore, this reduction in DA neuron counts strongly correlates with the decline in climbing ability. The decrease in both DA neuron counts and locomotor function from 1g to 3g/3gCO2 stressors resembles patterns observed in Drosophila models of Parkinson’s disease. Additionally, we detected increased apoptosis in hypergravity conditions (1.2g and 3g) in both males and females, further supporting the detrimental effects of these stressors on neuronal integrity. In addition to behavioral assessments, bioenergetic assays were conducted to evaluate various parameters, including mitochondrial membrane potential (MMP), adenosine triphosphate (ATP) quantification, Reactive Oxygen Species (ROS) levels, and mitotracker (indicative of mitochondrial abundance). Preliminary findings from these assays indicate notable alterations in ROS levels, mitochondrial membrane potential, and ATP production. Specifically, increased ROS levels suggest heightened oxidative stress within the cellular environment. Additionally, changes in mitochondrial membrane indicate potential disruptions in mitochondrial function, which can impact cellular energy metabolism and overall cellular health. Moreover, variations in ATP production suggest alterations in cellular energy production pathways, which may reflect adaptive responses to the imposed stressors. Further analysis and validation of these findings will provide deeper insights into the cellular responses to altered gravity and elevated CO2 levels. Understanding the bioenergetic implications of these stressors is critical for elucidating their effects on cellular physiology and may inform strategies for mitigating potential adverse outcomes in organisms exposed to such environmental conditions.

Bibliography Type:	Description: (Last Updated: 04/08/2025)
Abstracts for Journals and Proceedings	Kulkarni S. "Sex-specific physiological and transcriptomic responses of CNS to combined effects of spaceflight stressors in Drosophila melanogaster." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Abstracts for Journals and Proceedings	Kulkarni S. "Impact of the ISS environment on CNS in Drosophila melanogaster." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024